

An ORID–6E Generative AI Learning Companion to Foster Reflective Inquiry Competence in Data Visualization Education

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Abstract: *Data visualization courses often emphasize tool operation, yet many learners struggle to justify visualization choices, interpret data meaningfully, and reflect on the reasoning behind decisions. Increased reliance on generative AI for ready-made answers can further weaken inquiry and reflective thinking without structured guidance. This proposal develops an AI-supported reflective inquiry learning environment that integrates the ORID focused conversation method (Objective–Reflective–Interpretive–Decisional) with the 6E inquiry cycle (Engage–Explore–Explain–Engineer–Enrich–Evaluate). A stage-aware GenAI learning companion scaffolds inquiry and reflection during Tableau analysis of authentic educational or industry datasets, while a learning analytics dashboard supports instructor monitoring of inquiry behaviors, chat interactions, and reflection depth. Using a mixed-methods design, the study examines learning outcomes, motivation, self-regulated learning, and ORID reflection quality, and analyzes behavioral patterns from interaction logs. The project contributes a transferable design model and practical implementation to support evidence-based interpretation and reflective decision-making.*

Keywords: ORID, 6E inquiry learning, generative AI learning companion, data visualization literacy, reflective inquiry competence

1. Introduction

Data visualization is increasingly central to data-driven decision making, yet many courses remain operation-focused. Students may produce dashboards but still struggle to justify design choices, express uncertainty, or connect patterns to defensible claims (Lee et al., 2017). This emphasis on procedure is problematic because effective visual analysis depends on explicit reasoning that links evidence to interpretation and decision making while making assumptions and alternatives visible (Shrinivasan & van Wijk, 2008). The rapid diffusion of generative AI further intensifies this gap: AI can accelerate production of code and narratives but may reduce productive struggle and metacognitive monitoring when used as a shortcut (Kasneci et al., 2023). The key instructional challenge is therefore to position AI as a scaffold that keeps inquiry and reflection central. This proposal addresses this need by developing a reflective inquiry learning environment that structures analysis with an inquiry cycle, deepens reflection through ORID questioning, and embeds a stage-aware GenAI learning companion to elicit evidence-based reasoning rather than provide final interpretations.

Accordingly, this study addresses the following research questions:

RQ1: How does the ORID–6E GenAI learning companion system support learners’ inquiry and reflective reasoning during Tableau-based visualization tasks?

RQ2: How do different learning scenarios (individual practice, peer review, and group project iterations) shape learners’ use of system scaffolds and their evidence-based visualization decisions?

RQ3: What interaction and reflection patterns can be identified from system logs, and how can these patterns inform iterative refinement of the companion prompts and learning analytics dashboard design?

2. Literature Review

The design integrates inquiry-based learning as an activity structure, ORID focused conversation as a reflection scaffold, and a process-oriented GenAI companion with learning analytics for instructor monitoring

2.1 Inquiry-based learning and the 6E cycle

Inquiry-based learning emphasizes learning through problem framing, investigation, explanation, and evaluation, supported by appropriate guidance (Lazonder & Harmsen, 2016). Inquiry cycles provide a structure that helps learners move from curiosity to evidence-based claims (Pedaste et al., 2015). The 6E inquiry cycle extends common learning-cycle approaches by explicitly integrating design/engineering and enrichment/transfer phases, which are crucial for applied data work where students must refine artifacts and apply insights to new contexts. In data visualization projects, the 6E sequence helps learners connect real-world problems to iterative analytic actions and evaluation.

2.2 ORID focused conversation and deep reflection

ORID (Objective–Reflective–Interpretive–Decisional) is a structured reflection method that guides learners from describing observable facts, to articulating reactions, to interpreting meaning, and finally to deciding actions (Stanfield, 2000). In technical project work, ORID prevents premature conclusions by requiring learners to distinguish evidence from interpretation. ORID also provides a consistent language for reflective prompts embedded within learning tasks.

2.3 Generative AI learning companions and learning analytics

LLMs in education offer benefits and risks: feedback personalization and engagement gains may be offset by overreliance and hallucinated explanations when use is uncritical (Kasneci et al., 2023; Labadze et al., 2023). Accordingly, the proposed AI companion is designed to support process rather than deliver finished products by prompting evidence use, stating assumptions, testing alternative explanations, and justifying choices, consistent with pedagogical agents as learning companions and social-cognitive supports (Kim & Baylor, 2016). Learning analytics will provide instructors with actionable indicators of progress and difficulty to enable timely intervention (Ifenthaler & Yau, 2020). Prior work also supports the feasibility of a generative-AI inquiry companion for graduate data visualization learning, motivating the present design (Hung & Lai, 2025).

3. Methodology

The proposed environment integrates ORID as the primary reflective cycle, with the 6E inquiry cycle embedded within each ORID stage as task-specific prompts. Learners work on authentic datasets in Tableau and complete iterative inquiry tasks, while the GenAI companion provides stage-aware scaffolding. The platform also logs learning traces (e.g., prompt usage, reflection entries, task progress) to support analytics-driven instruction.

3.1 ORID as main cycle with embedded 6E sub-cycle

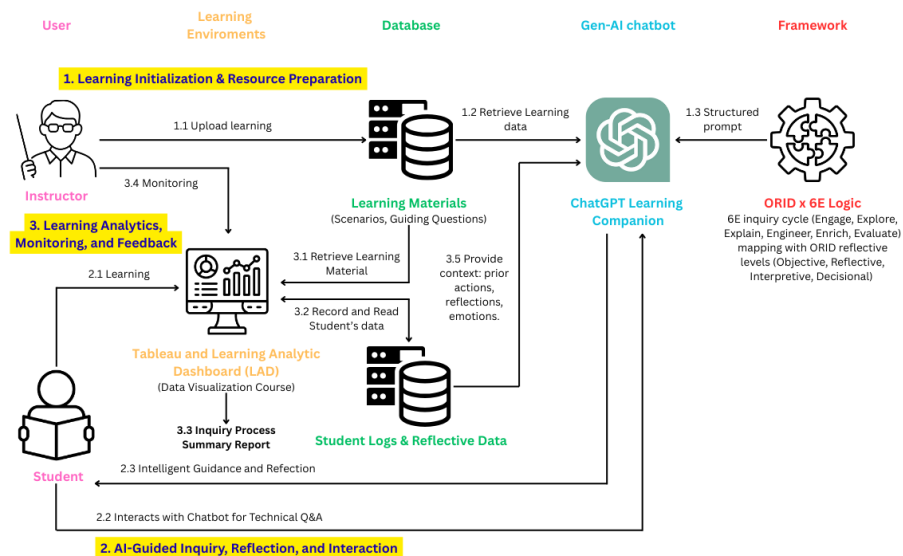
The learning flow begins with an ORID-based reflection frame. Within each ORID stage, students execute a mini 6E inquiry sequence appropriate to that stage. For example, during Objective (O), the companion triggers Engage/Explore prompts to help students notice patterns and define what is known from the visualization. During Reflective (R), prompts encourage learners to express uncertainty, difficulty, or surprise and to identify what students need to

investigate next. During Interpretive (I), the companion guides Explain/Engineer moves: constructing claims, testing alternatives, and refining visualization design to better support interpretations. Finally, during Decisional (D), the system prompts Evaluate/Enrich actions: validating conclusions, considering limitations, and planning transfer to a project deliverable or stakeholder decision.

3.2 System components

Figure 1 summarizes the system workflow in three stages with numbered steps. In Stage 1 (Learning Initialization & Resource Preparation), the instructor uploads learning resources (1.1), the system retrieves learning materials from the database (1.2), and the ORID × 6E framework generates structured prompts for the ChatGPT Learning Companion (1.3). In Stage 2 (AI-Guided Inquiry, Reflection, and Interaction), students conduct Tableau-based learning activities (2.1), interact with the chatbot for technical Q&A (2.2), and receive intelligent guidance and reflection scaffolding aligned with ORID × 6E (2.3). In Stage 3 (Learning Analytics, Monitoring, and Feedback), the LAD retrieves learning materials when needed (3.1), records and reads students' operational and reflective data (3.2), produces an inquiry process summary report (3.3), supports instructor monitoring (3.4), and provides the chatbot with learner context (e.g., prior actions, reflections, emotions) for more adaptive prompting (3.5).

Figure 1: System Functional Architecture Diagram



3.3 Prompting and guardrails

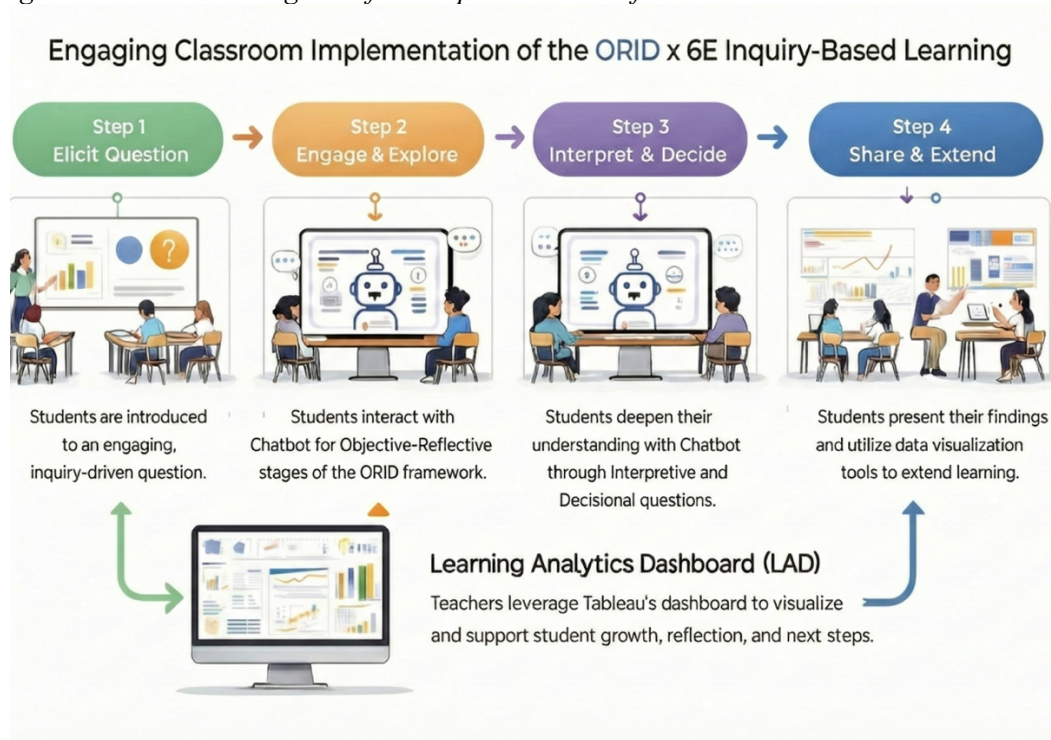
Companion prompts prioritize process rather than answers. The companion avoids producing full solutions and instead requests the chart elements or data evidence used, an interpretation stated as a claim with uncertainty, at least one alternative explanation, and a next-action decision. When a request seeks direct answers, the companion redirects the dialogue toward ORID steps (e.g., “What do you observe?” before “What does a pattern mean?”). This aligns with guidance research showing that inquiry benefits from structured support rather than unguided discovery (Lazonder & Harmsen, 2016).

3.4 Context and participants

The study will be implemented in a graduate Tableau-based data visualization course. Students will complete multiple project modules using authentic datasets and submit dashboards and ORID reflection logs. Participation will follow institutional ethics procedures, and all data will be de-identified. Figure 2 summarizes the implementation: each class begins with an instructor-led scenario, followed by GenAI-guided ORID inquiry. Objective prompts elicit descriptions of trends, differences, and anomalies; Reflective prompts elicit curiosity and initial hypotheses during exploration. During filtering and comparison, the companion prompts

question refinement and evidence-based reasoning. Interactions are logged and summarized in the learning analytics dashboard to show O–R–I–D coverage and inquiry depth. Interpretive prompts scaffold explanations of relationships and context, and Decisional prompts scaffold actionable recommendations through structured argumentation.

Figure 2: Schematic Diagram of the Implementation of the ORID-6E-AI Classroom Process



3.5 Quantitative measures and analysis (RQ1)

Pre/post assessments will measure changes in visualization literacy and related competencies. Visualization literacy can be assessed with VLAT or a course-aligned subset (Lee et al., 2017). Motivation and self-regulated learning can be measured with established instruments such as the Motivated Strategies for Learning Questionnaire (Pintrich et al., 1991) and self-regulation constructs aligned with social-cognitive perspectives (Zimmerman, 2000). Statistical analyses will compare pre/post changes and examine relationships with learning-process indicators.

3.6 Qualitative data and process analytics (RQ2–RQ3)

Three qualitative sources will be analyzed: ORID reflection logs, GenAI dialogue transcripts, and instructor observations and artifact reviews. Reflection quality will be coded using an ORID-aligned rubric, and themes will be developed through thematic analysis procedures (Braun & Clarke, 2006). In addition, interaction logs will support process modeling, such as identifying frequent transitions between inquiry actions and reflection states.

4. Expected Results

This project will deliver an ORID-as-backbone, 6E-embedded design model for reflective inquiry in Tableau-based data visualization learning, including a stage-aware GenAI prompting strategy that prioritizes evidence, assumptions, alternative explanations, and justification over answer generation. Future work will refine prompts and dashboard visualizations through iterative cycles informed by interaction logs, with the goal of enabling more adaptive scaffolding across individual practice, peer review, and group projects. Further development will expand learning analytics indicators to represent inquiry trajectories and reflection depth for instructor intervention and will test whether these indicators relate to changes in learning

outcomes, motivation, and self-regulated learning. Longer-term directions include evaluating transfer across datasets and courses, strengthening guardrails against overreliance and hallucinated reasoning, and releasing reusable prompt templates and dashboard specifications for broader adoption.

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